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Subject: Software Engineering

LAB 3: Architectural Justification

Student Club Event Ticketing & Budget Portal

Architecture Selection

I chose Layered Architecture for the Student Club Event Ticketing & Budget Portal.

The Layered Architecture is suitable for the system because it provides a clear separation between the user interface, business logic, and data management components. This makes the system easier to understand, develop, maintain, and modify.

Reason 1 – Separation of responsibilities:

The architecture separates different responsibilities into components such as Web/User Interface, Event & Proposal Management, Approval Workflow, Ticket & QR Management, and Database. This is particularly suitable for the portal because event proposal submission, multi-stage budget approval, event registration, and QR ticket validation are distinct functions that can be managed independently within the application.

Reason 2 – Simplicity and maintainability:

The portal is a moderate-sized application and does not require independently deployable services for every function. A layered design avoids the additional operational complexity and network communication overhead associated with Microservices Architecture while still providing a structured and maintainable system.

Security Advantage:

Role-Based Access Control (RBAC) can be implemented within the application/business layer so that each role, such as Club Lead, Faculty Coordinator, Finance Officer, Dean, Campus Admin, and Attendee, can access only the functions authorized for that role. This helps protect budget proposals, approval operations, event registrations, and ticket validation functions.

Performance Benefit:

The layered design allows frequently used operations to be organized efficiently within the application layer, while database access is handled separately. The Ticket & QR Management component can therefore perform QR ticket validation efficiently and support the system requirement of fast QR validation under the specified performance target.

Conclusion:

Overall, Layered Architecture provides an appropriate balance of simplicity, maintainability, security, and performance for the Student Club Event Ticketing & Budget Portal. It satisfies the functional requirements while avoiding the unnecessary complexity of a Microservices-based solution.